

Magma & LMFDB Mini-Course

Project B: Extending the LMFDB's Eisenstein Series Data (alpha)

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CTNT Summer School 2026

Why this project

The LMFDB classical modular forms database (<https://www.lmfdb.org/ModularForm/GL2/Q/holomorphic/>) tabulates *cuspidal* newforms with near-completeness up to large level/weight bounds. For *Eisenstein series* the story is more interesting:

- Eisenstein series are essentially **absent from the production (beta) version**. A 2013 issue on the LMFDB tracker ([#73](#)) flagged this: “surely some people would argue these are fundamental objects that should be in a database like this.”
- Eisenstein series **are present in the alpha version** (<https://alpha.lmfdb.org/ModularForm/GL2/Q/holomorphic/>). Data has been uploaded covering a range of levels, weights, and characters. Browse the alpha site and use the search interface to see what's there.

This makes the project quite different from contributing from a blank slate: the **schema is settled, a workflow exists, and existing data on alpha provides a template** for new entries. Your contribution – extending the coverage, filling specific gaps, adding cross-references – joins an active dataset rather than starting one.

Step 0: Read what's there

Before computing anything, spend the first hour exploring:

1. Open <https://alpha.lmfdb.org/ModularForm/GL2/Q/holomorphic/> and look for Eisenstein-related search options or filters. Note which level/weight/character ranges already have Eisenstein data and which fields are populated per entry.
2. Pull up several existing Eisenstein entries on alpha (try low levels and small weights first – e.g. E_4 , E_6 at level 1; Eisenstein series at level 5 or 7 with non-trivial character) and look at the displayed invariants and q -expansions.
3. Find the schema documentation on the alpha LMFDB (<https://alpha.lmfdb.org/api/> provides field-level documentation for every table). Identify the table name for Eisenstein series and read off its column structure. Take notes on:
 - how an Eisenstein newform is labeled,
 - how the indexing character pair (χ_1, χ_2) is recorded,
 - what auxiliary data (Hecke eigenvalues, L -function data, etc.) is attached.

4. Read [LMFDB issue #73](#) for historical context and any pinned discussion of the current state of Eisenstein coverage.

By the end of Step 0 you should be able to answer: which (N, k, χ) triples are already covered on alpha, which are not, and what data is required per entry.

The mathematics, briefly

For context as you compute: an **Eisenstein newform** at level N with character χ and weight k is indexed by a pair (χ_1, χ_2) of primitive Dirichlet characters with conductors N_1, N_2 satisfying:

- $\chi_1 \chi_2 = \chi$ (the product equals the nebentypus);
- $N_1 N_2 \mid N$;
- $\chi_1 \chi_2(-1) = (-1)^k$ (parity matches the weight);
- Not both χ_1, χ_2 trivial when $k = 2$ (to avoid E_2 , which is only quasimodular).

The Eisenstein series $E_k^{(\chi_1, \chi_2)}$ has explicit q -expansion (Diamond–Shurman, Theorem 4.5.2; Stein, Theorem 5.8):

$$E_k^{(\chi_1, \chi_2)}(q) = c_0 + \sum_{n \geq 1} \left(\sum_{d \mid n} \chi_1(n/d) \chi_2(d) d^{k-1} \right) q^n,$$

with c_0 a specific constant in terms of Bernoulli numbers and special L -values. The Hecke eigenvalues are determined explicitly by χ_1, χ_2, k, p .

Three phases

Phase 1 – Audit a slice of the alpha coverage and fill its gaps (4–8 hours).

Pick a manageable region of (N, k, χ) space – say, all (N, k) with $N \leq 50$ and $k \in \{2, 3, 4, 5\}$, varying over all characters $\chi \bmod N$. Then:

1. For each (N, k, χ) in your region, predict $\dim E_k(\Gamma_1(N), \chi)$ using the Cohen–Oesterlé formula (see Diamond–Shurman, Theorem 5.7.1, or *Computing Classical Modular Forms*, Theorem 4.3.4). Cross-check against `Dimension(EisensteinSubspace(ModularForms(...)))` in Magma or Sage.
2. Look up the corresponding alpha entries. For each (N, k, χ) where data is already present, verify that the dimension agrees with your formula and that the listed q -expansions are correct.
3. For each (N, k, χ) where data is missing or incomplete, compute the Eisenstein basis explicitly using the indexing (χ_1, χ_2) above, format the data per the alpha schema, and prepare it as a contribution.

This phase makes you both a *user* of the alpha data (catching any errors) and a *contributor* (filling in gaps you found).

Phase 2 – Substantive extension (10–20 hours).

Choose a direction where the alpha coverage thins out, and push it further. Some possibilities:

- **Larger levels at small weights.** Extend the existing coverage to $N \leq 1000$ at weight $k = 2$, ensuring the indexing (χ_1, χ_2) is recorded correctly.
- **Higher weights at small levels.** Extend to weights k up to 20 for $N \leq 100$.
- **Weight 1.** Eisenstein series of weight 1 are subtler (the Cohen–Oesterlé dimension formula does not apply; one uses Buzzard’s algorithm or direct construction from Hecke characters). If alpha is missing weight-1 data, this is a great target.
- **L -function data.** For each Eisenstein newform $f = E_k^{(\chi_1, \chi_2)}$, the L -function factors as $L(f, s) = L(\chi_1, s)L(\chi_2, s - k + 1)$. Check whether this factorization is recorded on the alpha entry; if not, add it (this is a one-line cross-reference, but very pedagogically useful for users).

Phase 3 – Cross-references and knowledge entries (after CTNT).

Eisenstein series carry deep structural information that current LMFDB pages don’t always expose:

- **Theta series of even unimodular lattices** live in $M_{r/2}(\mathrm{SL}_2(\mathbb{Z}))$ and decompose into Eisenstein and cuspidal parts. For each Niemeier lattice (cf. Project A), $\theta_\Lambda = aE_4^3 + b\Delta$ for explicit integers a, b that encode the absence/presence of roots. Add these explicit decompositions as cross-references between the lattice page and the Eisenstein-series page (this is one place where Projects A and B meet).
- **Galois representations.** The Galois representation attached to an Eisenstein newform is reducible – a direct sum of two 1-dimensional Galois characters from χ_1 and χ_2 . This is the simplest case of “modularity” and provides an excellent pedagogical pre-cursor to the 2-dim irreducible case from Lecture 4. Document the reducible Galois representation explicitly on each Eisenstein-newform page.
- **Hecke action.** Eisenstein series are Hecke eigenforms; their eigenvalues are determined explicitly by χ_1, χ_2, k, p . Verify the formulas on alpha and document them in knowledge entries.
- **Knowledge entries.** The existing entry at <https://www.lmfdb.org/knowledge/show/cmf.eisenstein> covers only the level-1 case. The general theory (Eisenstein newforms indexed by character pairs, the dimension formula, the Cohen–Oesterlé proof) deserves its own knowledge entry; if no such entry exists yet, write one.

Resources

- **Alpha LMFDB classical modular forms:** <https://alpha.lmfdb.org/ModularForm/GL2/Q/holomorphic/>.
- **Alpha API** (field-level documentation): <https://alpha.lmfdb.org/api/>.
- **LMFDB issue #73** (historical context): <https://github.com/LMFDB/lmfdb/issues/73>.
- Diamond & Shurman, *A First Course in Modular Forms*, Springer GTM 228, 2005. Chapter 5: “Modular Forms and Eisenstein Series”; Theorem 5.8 gives the explicit q -expansion formula.
- Stein, *Modular Forms, A Computational Approach*, AMS GSM 79, 2007. Chapter 5 covers Eisenstein series with attention to computation. Available at <https://wstein.org/books/modform/modform/>.
- Cohen & Strömberg, *Modular Forms: A Classical Approach*, AMS GSM 179, 2017. Chapter 17 is the standard modern reference for Eisenstein series of arbitrary level.

- Best, Bober, Booker, Costa, Cremona, Derickx, Lowry-Duda, Lee, Roe, Sutherland, Voight, *Computing Classical Modular Forms*, arXiv:2002.04717. Theorem 4.3.4 gives explicit dimension formulas for Eisenstein subspaces.
- Sage: `EisensteinForms(Gamma1(N), k), ModularForms(...).eisenstein_subspace()`.
- Magma: `EisensteinSubspace(ModularForms(Gamma1(N), k))`.

Where to ask for help. The LMFDB modular forms editorial board: Eran Assaf, David Roe, Andrew Sutherland, and John Voight are good first contacts. The CMF documentation is at <https://www.lmfdb.org/ModularForm/GL2/Q/holomorphic/Source>; the alpha schema inventory is reachable from the alpha API page.